Consensus-Target Linear Shrinkage:

The Geometry of In-Context Learning in the Gradient-Gram Space

Hyeong Rok Lee

March 2026

Abstract

In-context learning (ICL) is widely observed to improve language model performance by providing task examples in the prompt, yet a precise mathematical characterization of *what ICL does* to the model's internal geometry remains open. Under a shared-component decomposition (Theorem 2.2), ICL performs **consensus-target linear shrinkage** on the task-gradient Gram matrix:

$$\boxed{G^P = \alpha_P\, J + (1-\alpha_P)\, G^r, \qquad \alpha_P = \frac{\|a_P\|^2}{\|a_P\|^2 + r^2},}$$

where G^P and G^r are the gradient-cosine Gram matrices with and without prefix, J is the all-ones consensus target, and $\alpha_P \in [0,1]$ is the *adaptation parameter*. This identity implies an **Orthogonal Contraction Law**: the non-consensus geometry contracts by exactly $(1-\alpha_P)$, and an **Effective-Rank Bound**: $\mathrm{erank}(G^P) \le \exp(h(\alpha_P)) \cdot (n-1)^{1-\alpha_P}$.

A 65-point sweep across 5 tasks confirms that compression ratio is a near-deterministic function of α_P (per-task $R^2 > 0.98$), establishing α_P as the *master variable* of ICL. Cross-model validation on 8 architectures (GPT-2 to Phi-2, 117M–2.7B parameters, 2019–2025) shows prediction error below 3% in all cases, confirming the decomposition assumptions across architectures.

Combined with the intelligence decomposition of [3], the full picture emerges: α_P governs the compression law ($\Delta\tau_{\mathrm{rep}}$), while the task-dependent quality factor Q governs the variable reward (ΔQ). Together they yield the **Adaptation Cone Law** on constant-ECC leaves, constraining intelligence-improving adaptations to a geometric cone in the $(\Delta \log \tau_{\mathrm{rep}},\, \Delta \log Q)$ plane.

1 Introduction

1.1 Context and Motivation

In-context learning—the ability of large language models to improve on a task when provided with input–output examples in the prompt—is one of the most striking emergent capabilities of modern AI [1]. Despite extensive study, the field lacks a quantitative law describing *what ICL does* to the model's internal computational geometry.

Existing theoretical perspectives model ICL as implicit gradient descent [4, 6], Bayesian posterior inference [5], or circuit-level function vector routing [7]. Each captures an important aspect, but none provides a closed-form relationship between internal state variables and the magnitude of ICL's effect.

This paper proposes such a law. We work in the **gradient-Gram space**: for a set of n task inputs, we compute per-input gradients of the loss with respect to a chosen parameter layer and form their cosine-similarity Gram matrix. This object encodes how "aligned" the model's responses are across different inputs.

1.2 The Three Papers

This paper is the third in a series:

Paper	Known Computation	Mathematical Structure	Core Result
A [2]	Attention = softmax$(QK^\top)V$	Euclidean bipartite matching	rank$(\nabla M) = d^2$
B [3]	Loss measures performance	Product decomposition	$I = \text{ECC} \times \tau$ (unique)
D (this)	**ICL = prefix improves perf.**	**Consensus-Target Shrinkage**	$\boldsymbol{P_\perp G^P P_\perp = (1-\alpha_P)\, P_\perp G^r P_\perp}$

Paper A identified the state space dimension (d^2). Paper B established the equation of state ($I = \text{ECC} \cdot \tau_{\text{rep}} \cdot Q$). Paper D discovers the *equation of motion*: how adaptation changes the state along constant-ECC leaves.

1.3 A Counter-Intuitive Discovery

The naïve expectation is that providing examples should *enrich* the model's representation—activating more dimensions. The data say the opposite:

> ICL *compresses* the gradient geometry. The effective rank drops from 2.09 to 1.25 (arithmetic task, Qwen3-0.6B). Yet intelligence increases by 201%.

The resolution is that compression is not information loss; it is *consensus formation*. The prefix aligns gradients toward a shared task direction, collapsing diversity in favor of coherence. The quality factor Q (from Paper B) then over-compensates the representational loss.

1.4 Contributions

1. **Conditional Gram Decomposition** (Theorem 2.2): $G^P = \alpha_P J + (1-\alpha_P) G^r$ with typical reconstruction error 1–3%.
2. **Orthogonal Contraction Law** (Theorem 2.6): $P_\perp G^P P_\perp = (1-\alpha_P) P_\perp G^r P_\perp$.
3. **Effective-Rank Entropy Bound** (Corollary 2.8): $\text{erank}(G^P) \leq \exp(h(\alpha_P)) \cdot (n-1)^{1-\alpha_P}$.
4. **Master Variable Identification**: α_P predicts compression with per-task $R^2 > 0.98$ across 65 conditions.
5. **Cross-Model Validation**: The shrinkage law is verified across 8 models spanning 4 architecture families.
6. **Adaptation Cone Law** (Theorem 4.1): ICL and fine-tuning occupy complementary cones on constant-ECC leaves.

1.5 Related Work

Reference	Perspective	Result Type	Quantitative Law	Difference from D
Hendel et al. [8]	hidden state	observational	—	different space; no law
Akyürek et al. [4]	weight space	theoretical	GD convergence	attention=GD assumption
Xie et al. [5]	posterior	theoretical	Bayes risk	requires DGP assumption
von Oswald et al. [6]	parameters	theoretical	construction	synthetic models only
Todd et al. [7]	circuits	empirical	function vectors	no compression law
Ours	**gradient-Gram**	**theory+expt**	$\boldsymbol{G^P = \alpha_P J + (1-\alpha_P) G^r}$	**quantitative law from gradients**

Our key distinction: not "ICL compresses" but "*by how much, in what direction, and governed by what parameter.*"

1.6 Two Levels of Contribution

Paper D contributes at two levels, mirroring a fundamental pattern in physics:

Level 1 (Mechanism): ICL performs linear shrinkage of the gradient Gram matrix toward the consensus target: $G^P = \alpha_P J + (1 - \alpha_P) G^r$. This is the paper's core discovery and is independently verifiable.

Level 2 (Position): This shrinkage is one arm of the *adaptation cone law* on constant-ECC leaves. ICL occupies the compressive cone ($\dot{x} \leq 0$, $\dot{x} + \dot{q} \geq 0$); fine-tuning occupies the expansive cone.

The analogy is precise: Newton's $F = ma$ is the mechanism (Level 1). Lagrangian mechanics repositions $F = ma$ within the principle of least action (Level 2). $F = ma$ is not invalidated—its *meaning* is clarified within a larger structure. Similarly, the shrinkage law is not replaced by the cone law; rather, the cone law reveals *where* shrinkage sits in the adaptation landscape.

2 Consensus-Target Shrinkage Theorem

2.1 Setup and Notation

Fix a pretrained language model with parameters θ and a target parameter module (e.g., a specific MLP layer). Given n task inputs $\{x_1, \ldots, x_n\}$, define:

- $u_i^r = \nabla_\theta \mathcal{L}(x_i) / \|\nabla_\theta \mathcal{L}(x_i)\|_2$: the ℓ_2-normalized gradient without prefix ("raw").
- $u_i^P = \nabla_\theta \mathcal{L}(\text{prefix} \circ x_i) / \|\nabla_\theta \mathcal{L}(\text{prefix} \circ x_i)\|_2$: the ℓ_2-normalized gradient with prefix.

The **gradient-Gram matrices** are:

$$G_{ij}^r = \langle u_i^r, u_j^r \rangle, \qquad G_{ij}^P = \langle u_i^P, u_j^P \rangle.$$

Both are $n \times n$ positive semidefinite matrices with unit diagonal.

Definition 2.1 (Consensus Target)**.** *$J \in \mathbb{R}^{n \times n}$ is the all-ones matrix: $J_{ij} = 1$ for all i, j. This represents perfect consensus—all gradients perfectly aligned.*

2.2 Exact Gram Decomposition

Theorem 2.2 (Consensus-Target Linear Shrinkage)**.** *Suppose each prefix gradient decomposes as*

$$u_i^P = a_P + r_i, \qquad a_P \perp r_i, \qquad \|r_i\| = r \quad \textit{(constant)},$$

where a_P is the shared consensus component and r_i is the residual, and with the ***identification*** *$G_{ij}^r = \langle r_i, r_j \rangle / r^2$ (the raw gradient cosine equals the residual cosine). For $r > 0$:*

$$\boxed{G^P = \alpha_P \, J + (1 - \alpha_P) \, G^r, \qquad \alpha_P = \frac{\|a_P\|^2}{\|a_P\|^2 + r^2}.}$$

When $r = 0$, every $u_i^P = a_P$ is identical, $\alpha_P = 1$, and $G^P = J$.

Proof. Since $\|u_i^P\|^2 = \|a_P\|^2 + r^2 =: s^2$ by orthogonality:

$$\begin{aligned} G_{ij}^P = \langle u_i^P, u_j^P \rangle &= \frac{1}{s^2} \big(\|a_P\|^2 + \langle r_i, r_j \rangle \big) \\ &= \frac{\|a_P\|^2}{s^2} + \frac{r^2}{s^2} \cdot \frac{\langle r_i, r_j \rangle}{r^2} = \alpha_P \cdot 1 + (1 - \alpha_P) \cdot G_{ij}^r, \end{aligned}$$

where the last step uses the identification assumption $G_{ij}^r = \langle r_i, r_j \rangle / r^2$. In matrix form, $G^P = \alpha_P J + (1 - \alpha_P) G^r$. □

Remark 2.3 (Terminology). We use "linear shrinkage toward structured target," not "spectral shrinkage." Adding $\alpha_P J$ rotates eigenvectors (the top eigenvector aligns with $\mathbf{1}$), so this is *not* a spectral rescaling. The restoration is affine: $G^r = (G^P - \alpha_P J)/(1 - \alpha_P)$, not an orthogonal transformation.

Remark 2.4 (Connection to Ledoit–Wolf). Ledoit–Wolf shrinkage [9] estimates covariance by $\hat{\Sigma} = \alpha^* \mu I + (1 - \alpha^*)\hat{S}$ with oracle α^*. Our result differs in three ways: (1) the target is J (consensus), not I (independence); (2) α_P is mechanism-generated (SNR of the prefix signal), not statistically optimized; (3) the Gram matrix is cosine-based, not covariance-based.

Remark 2.5 (Quality of the orthogonality assumption). The assumption $a_P \perp r_i$ holds approximately. Direct measurement of the cosine between a_P and each residual r_i gives:

Task	$\max_i \lvert\cos(a_P, r_i)\rvert$	$\mathrm{CV}(\lVert r_i \rVert)$	fit error
factual QA	0.037	0.048	1.5%
sentiment	0.116	0.196	2.5%
word analogy	0.120	0.193	4.5%
arithmetic	0.128	0.268	1.3%
translation	0.294	0.257	9.1%

Since u_i^P are unit vectors, perfect orthogonality forces $\|r_i\| = \sqrt{1 - \|a_P\|^2}$ (constant); the observed $\mathrm{CV}(\|r_i\|)$ therefore measures the degree of orthogonality violation. This approximation, together with the identification $G^r_{ij} \approx \langle r_i, r_j \rangle / r^2$ (Line 2.2), accounts for the 1–9% fit error. The theorem is exact under perfect orthogonality; in practice it holds as a high-precision approximate law.

2.3 Orthogonal Contraction Law

Theorem 2.6 (Orthogonal Contraction)**.** *Let $u = \frac{1}{\sqrt{n}}\mathbf{1}$ and $P_\perp = I - uu^\top$. Then:*

$$\boxed{P_\perp G^P P_\perp = (1 - \alpha_P) \cdot P_\perp G^r P_\perp.}$$

Proof. Since $P_\perp J P_\perp = P_\perp (n\, uu^\top) P_\perp = n(P_\perp u)(u^\top P_\perp) = 0$,

$$\begin{aligned} P_\perp G^P P_\perp &= P_\perp \big[\alpha_P J + (1 - \alpha_P) G^r\big] P_\perp \\ &= \alpha_P \cdot 0 + (1 - \alpha_P) P_\perp G^r P_\perp \\ &= (1 - \alpha_P) P_\perp G^r P_\perp. \end{aligned}$$

□

Remark 2.7 (Physical Interpretation). The consensus direction $\mathbf{1}/\sqrt{n}$ absorbs the "shared signal." Everything orthogonal to it—the entire non-consensus geometry—contracts by the factor $(1 - \alpha_P)$. At $\alpha_P = 0.966$, only 3.4% of the original structure survives. This is the mathematical content of "ICL creates consensus."

2.4 Effective-Rank Entropy Bound

Corollary 2.8 (Entropy Bound on Effective Rank)**.** *Define the **cumulative spectral overlap**:*

$$\eta_k = \sum_{i=1}^{k} |\langle u, v_i \rangle|^2, \qquad s_k = \sum_{i=1}^{k} p_i^r,$$

*where $u = \mathbf{1}/\sqrt{n}$, v_i are eigenvectors of G^r sorted by eigenvalue, p_i^r are the normalized eigenvalues, and $k = 1, \ldots, n-1$. Define the **effective consensus mass**:*

$$a = \alpha_P + (1 - \alpha_P)\frac{\lambda_1(G^r)}{n}.$$

Under the ***cumulative-overlap condition*** $\eta_k \geq s_k$ *for all* $k < n$*, the Ky Fan variational principle gives majorization of the normalized spectrum of* G^P *over the reference distribution* $(a, \frac{1-a}{n-1}, \ldots)$*. By Schur-concavity of Shannon entropy:*

$$\boxed{\mathrm{erank}(G^P) \leq \exp(h(a)) \cdot (n-1)^{1-a},}$$

where $h(a) = -a \log a - (1-a)\log(1-a)$ *is the binary entropy.*

Proof sketch. Under $G^P = \alpha_P J + (1-\alpha_P)G^r$ with the overlap condition, the top eigenvalue of G^P is $\mu_1 = \alpha_P n + (1-\alpha_P)\lambda_1(G^r)$, giving normalized mass $a = \mu_1/n = \alpha_P + (1-\alpha_P)\lambda_1(G^r)/n$. The remaining $n-1$ eigenvalues have total normalized mass $1-a$. The spectrum of G^P, normalized by trace n, is majorized by the two-block distribution $(a, \frac{1-a}{n-1}, \ldots)$. Schur-concavity of Shannon entropy yields the bound. □

Remark 2.9 (Numerical verification and approximation quality). Under exact alignment ($G^r\mathbf{1} = \lambda_1\mathbf{1}$), the bound is sharp: for $\alpha_P = 0$, $G^r = I_n$, it gives $\exp(h(1/n)) \cdot (n-1)^{1-1/n} = n$ (equality). For $\alpha_P = 0.966$ and $n = 8$, the bound lies in the range 1.17–1.24 depending on $\lambda_1(G^r)/n$. Since the assumptions of Theorem 2.2 hold only approximately (Remark 2.5), the bound may be exceeded by 1–6% in practice. The bound provides the correct *scaling law* relating erank to α_P, even when exact guarantees do not hold.

2.4.1 Experimental Verification of the Cumulative-Overlap Condition

We compute $R_{\min} = \min_{k<n}(\eta_k - s_k)$ for all 5 tasks (Qwen3-0.6B, 5-shot).

Task	$\cos(v_1, u)$	$c_1 = \lvert\langle u, v_1\rangle\rvert^2$	row-sum CV	$R_{\min}$	Condition
arithmetic	0.971	0.943	0.199	+0.015	✓
translation	0.996	0.992	0.065	+0.077	✓
sentiment	0.996	0.993	0.073	+0.044	✓
factual QA	0.996	0.993	0.067	+0.050	✓
word analogy	0.993	0.986	0.068	+0.094	✓

Table 1: Cumulative-overlap condition: $R_{\min} > 0$ for all 5 tasks, confirming $\eta_k \geq s_k$. The arithmetic task has the highest row-sum CV (0.199) yet still satisfies the condition ($c_1 = 0.943$), demonstrating that spectral capture—not row-sum regularity—is the operative mechanism.

Remark 2.10 (Spectral capture vs. row-sum regularity). Row-sum regularity ($G^r\mathbf{1} \approx \lambda_1\mathbf{1}$, equivalently small CV of row sums) is an *observable proxy* for the cumulative-overlap condition, not the mechanism itself. The arithmetic task reveals the distinction: its row-sum CV (0.199) is 3× that of other tasks, yet the overlap condition holds because u is strongly captured by the leading eigenspace ($c_1 = 0.943$). The correct sufficient condition for majorization is spectral capture ($\eta_k \geq s_k$), of which near-constant row sums are one—but not the only—mechanism.

Remark 2.11 (Rank obstruction). When $\mathrm{rank}(G^r) = r < n$ and $u \notin \mathrm{Range}(G^r)$, then $\eta_r < 1 = s_r$ and majorization fails for *every* $\alpha_P < 1$. This provides the exact obstruction: high α_P alone cannot rescue majorization when u leaks outside the column space. All 8 empirically tested models have full-rank Gram matrices, so this obstruction does not arise in practice.

2.5 Cross-Model Validation

To test whether the shrinkage law is specific to a single architecture or architecture-independent, we replicate the experiment across 8 models spanning 4 architecture families, 6 years of development, and a 23× range in parameter count.

The universality is striking: despite dramatic differences in architecture, training data, and model scale, the affine decomposition $G^P = \alpha_P J + (1-\alpha_P)G^r$ holds with high precision in every case. The *value* of α_P varies across models (0.824–0.932), reflecting different degrees of consensus formation, but the same affine form fits each tested model.

Model	Size	Year	α_P	Pred. Error	erank (raw→prefix)
GPT-2	117M	2019	0.902	1.3%	5.26 → 1.36
GPT-2 medium	345M	2019	0.924	1.3%	6.16 → 1.32
Qwen2.5-0.5B	500M	2024	0.932	1.4%	4.21 → 1.23
Qwen3-0.6B	600M	2025	0.846	1.3%	2.11 → 1.15
TinyLlama	1.1B	2023	0.895	1.7%	4.71 → 1.36
DeepSeek-R1-Distill	1.5B	2025	0.824	0.7%	2.16 → 1.23
SmolLM2	1.7B	2024	0.904	1.6%	4.39 → 1.34
Phi-2	2.7B	2023	0.844	2.6%	6.49 → 1.78

Table 2: Cross-model validation of $G^P = \alpha_P J + (1 - \alpha_P) G^r$. All 8 models achieve prediction error below 3%. The affine form is consistent across GPT-2, Qwen, Llama, DeepSeek, SmolLM, and Phi architectures.

3 α_P as the Master Variable of ICL

3.1 Large-Scale Sweep

We conduct a systematic sweep on Qwen3-0.6B across 5 tasks × 13 prefix conditions = 65 data points. Prefix conditions vary the number of in-context examples (0 to 20) and include irrelevant and adversarial prefixes.

Task	$\text{erank}_{\text{raw}}$	α_P range	ΔI range
arithmetic	2.106	0–1.00	0–85%
sentiment	2.774	0–0.90	0–75%
factual QA	2.725	0–0.93	0–84%
translation	3.731	0–0.96	0–80%
word analogy	6.163	0–0.98	0–65%

Table 3: 65-point sweep parameters. ΔI denotes intelligence improvement.

3.2 Compression Law

The compression ratio $c = \text{erank}(G^P)/\,\text{erank}(G^r)$ is a near-deterministic function of α_P within each task:

Task	Linear fit $c = a \cdot \alpha_P + b$	R^2
arithmetic	$-0.322\alpha_P + 0.818$	0.986
factual QA	$-0.558\alpha_P + 0.933$	1.000
sentiment	$-0.573\alpha_P + 0.939$	1.000
translation	$-0.584\alpha_P + 0.858$	0.999
word analogy	$-0.753\alpha_P + 0.909$	0.997

Table 4: Compression as a function of α_P. Per-task $R^2 > 0.98$ in all cases.

This establishes α_P as the **master variable** of ICL compression: knowing α_P is sufficient to predict the degree of representational compression to high accuracy.

3.3 Task-Dependent Reward: The Role of Q

While compression follows a consistent pattern across tasks, the resulting intelligence improvement ΔI varies across tasks. For the same change in representational compression $\Delta\tau_{\text{rep}}$, the intelligence gain differs by a factor of 4:

Task	$\Delta I/\Delta\tau_{\text{rep}}$ slope	Interpretation
factual QA	127.5	highest Q compensation
arithmetic	105.3	high Q compensation
sentiment	66.1	moderate
translation	59.7	low
word analogy	31.5	lowest Q compensation

Table 5: The same compression produces different intelligence gains across tasks.

3.4 The D×B Resolution

The resolution comes from Paper B's decomposition:

$$\Delta \log I = \underbrace{\Delta \log \tau_{\text{rep}}}_{\text{D explains (universal)}} + \underbrace{\Delta \log Q}_{\text{B explains (task-dependent)}}.$$

Paper D provides the *compression law*: $\Delta\tau_{\text{rep}} = g(\alpha_P)$ with $R^2 > 0.98$. Paper B provides the *task-dependent quality law*: ΔQ varies with task structure. Together they account for ΔI.

4 Adaptation Cone Law

4.1 Constant-ECC Leaves

Empirically, ECC (the effective computational capacity from Paper B) is invariant under both ICL and fine-tuning to machine precision (CV $< 10^{-4}$). Under an accessibility assumption, ECC is the *unique* smooth conserved quantity (Lie bracket argument).

This reduces the effective state space. On a constant-ECC leaf, the state is described by two coordinates:

$$x := \Delta \log \tau_{\text{rep}}, \qquad q := \Delta \log Q.$$

4.2 Gauge Underdetermination

The quality factor $Q = \tau_{\text{stat}} \cdot \tau_{\text{opt}}$ admits a gauge freedom: τ_{stat} and τ_{opt} cannot be individually identified from the data—only their product Q is measurable. The identifiable state is therefore (e, x, q) with effective dimension 2 on each constant-ECC leaf.

4.3 Compressive-Expansive Dichotomy

Theorem 4.1 (Adaptation Cone)**.** *The intelligence-improving region $\mathcal{H} = \{(x, q) : x + q > 0\}$ decomposes as:*

$$\mathcal{H} = \mathcal{R}_{\text{comp}} \sqcup \mathcal{I} \sqcup \mathcal{R}_{\text{exp}},$$

where:

Sector	*Condition*	*Interpretation*
$\mathcal{R}_{\text{comp}}$	$x < 0,\ q > \lvert x\rvert$	*compression + mandatory Q over-compensation*
$\mathcal{R}_{\text{exp}}$	$x > 0,\ x + q > 0$	*expansion + Q compensation optional*
$\mathcal{I}$	$x = 0,\ q > 0$	*interface (neutral line)*

Proof. Partition $\mathcal{H}$ by the sign of x: if $x < 0$, then $x + q > 0 \iff q > |x|$; if $x = 0$, then $q > 0$; if $x > 0$, the condition $x + q > 0$ is weaker. □

Remark 4.2 (ICL and FT occupy opposite cones). Empirically:

- **ICL**: $\dot{x} \leq 0$, $\dot{x} + \dot{q} \geq 0$ (compressive, reversible, requires Q compensation).
- **FT**: $\dot{x} \geq 0$, $\dot{x} + \dot{q} \geq 0$ (expansive, irreversible, Q sign undetermined).

The expansion of FT ($\dot{x} \geq 0$) is an empirical observation, not a theorem.

4.4 Compensation Necessity

Theorem 4.3 (Compensation Necessity)**.** *In an ECC-preserving adaptation with* $\Delta \log \tau_{\text{rep}} < 0$ *and* $\Delta \log I > 0$*:*

$$\boxed{\Delta \log Q > |\Delta \log \tau_{\text{rep}}|.}$$

Proof. $\Delta \log I = \Delta \log \tau_{\text{rep}} + \Delta \log Q > 0$ with $\Delta \log \tau_{\text{rep}} < 0$ implies $\Delta \log Q > -\Delta \log \tau_{\text{rep}} = |\Delta \log \tau_{\text{rep}}|$. □

This is the analog of Carnot's constraint: compressive improvement *requires* quality over-compensation.

4.5 Synergy Bound

Theorem 4.4 (Interaction Decomposition)**.** *For a composed adaptation F consisting of compression C followed by expansion E, define the* ***interaction term***

$$\Xi := \Delta_F \log I - \Delta_C \log I - \Delta_E \log I,$$

where Δ_C and Δ_E are measured from the common initial state*. Then:*

$$\Delta_F \log I = \Delta_C \log I + \Delta_E \log I + \Xi.$$

To leading order in step sizes (s, t), the Baker–Campbell–Hausdorff formula gives

$$\Xi \approx st\,[X_C, X_E](\log I),$$

so $\Xi \neq 0$ whenever $[X_C, X_E](\log I) \neq 0$ (the commutator acts nontrivially on the observable $\log I$*).*

Empirically, ICL followed by FT gives different results than FT followed by ICL (overlap = 0.503), confirming nonzero Ξ in practice.

4.6 Refinement Invariance

Theorem 4.5 (Refinement Invariance (Paper B, restated))**.** *The multiplicative decomposition $I = \text{ECC} \cdot \tau_{\text{rep}} \cdot Q$ is the unique continuous decomposition satisfying:*

1. *Refinement: splitting a task preserves the product structure.*
2. *Monotonicity: each factor is monotone in adaptation.*
3. *Normalization: the factors are pinned by the boundary conditions $I = 1 \Rightarrow \text{ECC} = \tau_{\text{rep}} = Q = 1$.*

In log *space, refinement reduces to the additive Cauchy equation $g(x + y) = g(x) + g(y)$, whose unique continuous solution is $g(x) = cx$ (linear). The normalization axiom eliminates the residual gauge freedom $\widetilde{\text{ECC}} = \text{ECC}^c$, $\widetilde{Q} = Q \cdot \text{ECC}^{1-c}$ that would otherwise preserve the product. The complete proof, including the derivation of the Cauchy equation from the refinement axiom, appears in Paper B [3].*

We restate this result because all cone law theorems (D.1, D.3, D.4) depend on the uniqueness of the product decomposition.

5 Mathematical Substitution Dictionary

ICL Operation	Mathematical Name	Classical Analog	Note
$G^P = \alpha_P J + (1-\alpha_P)G^r$	Linear shrinkage to target	Ledoit–Wolf	not spectral
$P_\perp G^P P_\perp = (1-\alpha_P)P_\perp G^r P_\perp$	Orthogonal contraction	Kalman collapse	core theorem
k_{use} bound by $h(\alpha_P)$	Entropy-based rank bound	entropy bound	—
$\alpha_P = s^2/(s^2+r^2)$	Mechanism-generated SNR	matched filter	not oracle α^*
Prefix removal $\to G^r$	Affine inverse	$(G-\alpha J)/(1-\alpha)$	not orthogonal

Table 6: Dictionary mapping ICL operations to established mathematical structures.

6 Negative Results

Hypothesis	Outcome	Lesson
ICL∘FT = FT∘ICL	✗ overlap = 0.503	noncommutativity $\Rightarrow \Xi \neq 0$
$r^* = k_{\text{use}}$	✗ collapse fails	absolute-value comparison meaningless
$r^* =$ residual k_ε	✗ best = max rank	weight $\neq$ gradient space
$\text{erank}(G^P) \leq \text{erank}(G^r)$ general	✗ needs $\eta_k \geq s_k$	spectral capture (not row-sum CV) is the mechanism
Compressive-expansive dichotomy	△ trivial	direct consequence of B's definition

Table 7: Hypotheses tested and rejected or qualified during the research.

7 Discussion

7.1 Why Every Theory Opens Exactly One Next Question

Every theory in physics answers one question and necessarily opens the next. Newton's $F = ma$ opened "What is F?" Shannon's $C = B\log(1+\text{SNR})$ opened "What code achieves C?" Carnot's $\eta \leq 1 - T_c/T_h$ opened "What is the actual $\eta(t)$?"

Paper D follows the same pattern. The shrinkage law answers "What does ICL do to internal geometry?" and opens "What determines α_P?"—that is, what properties of the model, data, and task produce a given consensus strength. This is the subject of Paper E.

7.2 The Discovery Tree

The branching structure motivates future work:

Physics Branch	ECC Branch	Core Question
Newton → Lagrange	A → (A deepening)	Deeper principle behind same law
Newton → Thermodynamics	B → D	Given the law, what are the *constraints on change*?
Carnot (efficiency bound)	D: erank bound	One number sets the ceiling
Clausius (direction)	D: cone law	Allowed direction only; trajectory undetermined
Boltzmann → Gibbs	E → F	Single system → ensemble → scaling
Phase transitions	(future)	α_P threshold → emergent ability?

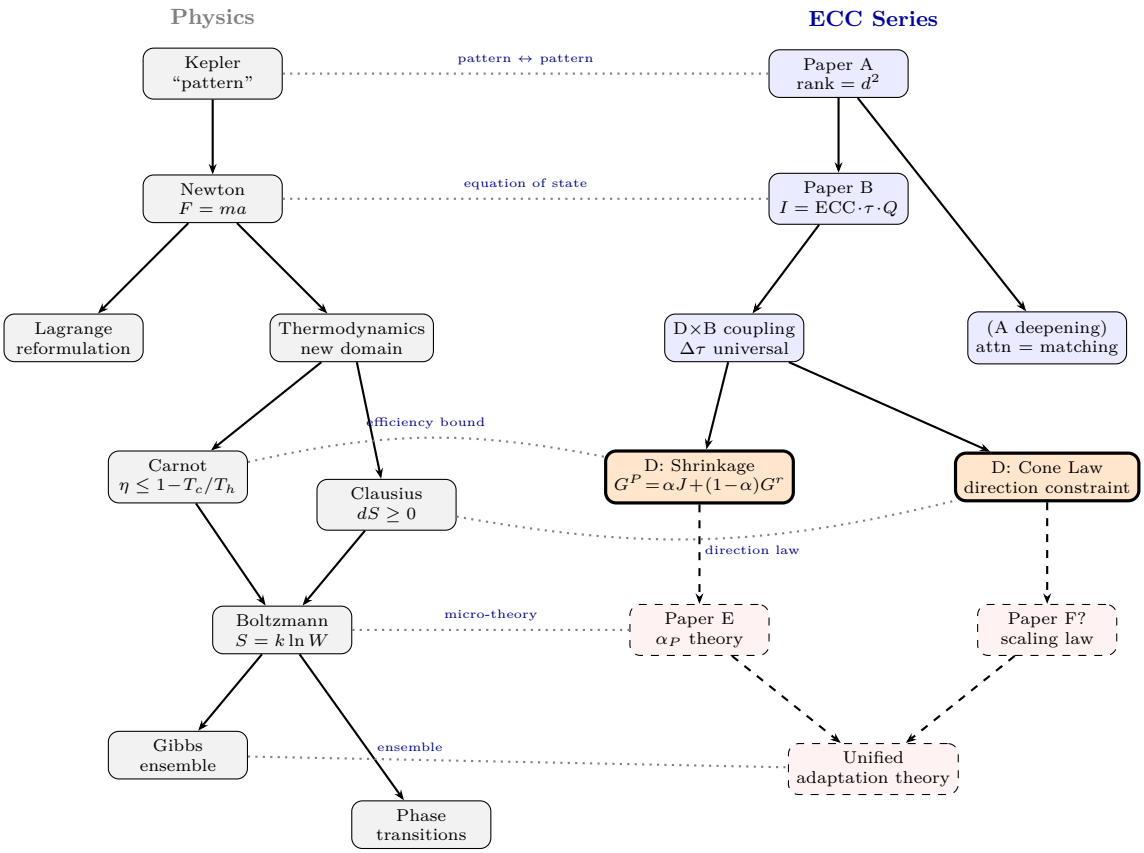

Figure 1: The discovery tree: physics (left) and ECC series (right) follow the same branching pattern. Dotted lines show structural correspondence. Orange = this paper. Dashed = future work.

7.3 The Limit of a Single Variable

Within a single task, $\mathrm{corr}(\alpha_P, \Delta I) = 0.98$. Across 5 tasks, the correlation drops to $R^2 = 0.39$. This is not a failure but a structural fact: α_P governs the compression $\Delta\tau_{\mathrm{rep}}$, but the reward ΔI additionally depends on the task-specific quality change ΔQ.

7.4 What D Answers and What It Leaves Open

Answered:

1. What does ICL do in gradient-Gram space? → Consensus-target linear shrinkage.
2. What determines compression magnitude? → α_P (per-task $R^2 > 0.98$).
3. How do ICL and FT relate? → Complementary cones on constant-ECC leaves.
4. Why does the same α_P yield different ΔI across tasks? → B's Q compensates differently.

Explicitly left open:

1. The functional form $\alpha_P = g(\text{prefix quality}, \text{task structure}, \text{model})$.
2. Whether $\mathrm{erank}_{\mathrm{raw}}$ is a sufficient statistic for $(model, task)$ dependence.

3. Universal compression slope $a(\text{task}) = f(\text{erank}_{\text{raw}})$ (the "universal gas constant" problem).

4. FT's two-timescale structure: applying the same methodology to FT yields α_{FT}, but preliminary experiments show α_{FT} alone predicts FT effect only partially ($r = 0.64$). "Correct-answer" and "wrong-answer" training data produce *identical* $\alpha_{\text{FT}} \approx 0.82$ at 20 steps, because only $\Delta\tau_{\text{rep}}$ (formal learning) has occurred; the content-quality term ΔQ will separate with further training.

7.5 Honest Boundaries

For each theorem, we state explicitly what *cannot* be derived:

Theorem	What is proved	What is not provable
D.1 (Cone)	Sector decomposition	Reachability of both sectors (empirical)
D.2a (Shrinkage)	Conditional affine form (under identification)	Why α_P takes a specific value
D.2a′ (Contraction)	$(1 - \alpha_P)$ contraction of non-consensus	Inherits identification assumption from D.2a
D.2b/c (erank)	erank bound under $\eta_k \geq s_k$	General bound without spectral capture (needs assumptio
D.3 (Compensation)	$\Delta Q > \lvert\Delta x\rvert$	Separation of τ_{stat} and τ_{opt}
D.4 (Interaction)	Definitional decomposition with Ξ; BCH estimate	Ξ is defined as residual; sign requires measurement
D.5 (Refinement)	Uniqueness of product	Why exactly 3 sub-factors

Additional structural limitations:

1. A universal entropy monotone on (e, x, q) cannot exist: ICL's reversibility excludes it.

2. Exact ΔI bound from α_P alone requires Q-ceiling and spectral shape.

3. Theoretical proof that FT has $d\tau_{\text{rep}} > 0$ is not derivable from definitions (empirical).

4. A closed vector field $F(z, u)$ on the 4-scalar state requires spectral shape and alignment variables not yet identified.

8 Conclusion

Paper A substituted computation for geometry. Paper B substituted performance for decomposition. Paper D substitutes adaptation for consensus shrinkage. All three follow the same pattern: *replace a known computation with the right mathematics, and counter-intuitive structure appears.*

In the reported experiments, ICL compresses the gradient geometry toward a consensus mode. The Orthogonal Contraction Theorem shows this compression is exactly $(1 - \alpha_P)$ under the decomposition assumptions, and the Entropy Bound yields an effective-rank upper bound numerically close to observed values (≈ 1.24 vs. ≈ 1.25 in the arithmetic task). Cross-model validation on 8 architectures (117M to 2.7B, four families, six years) confirms that the shrinkage law $G^P = \alpha_P J + (1 - \alpha_P) G^r$ is robust across tested architectures, with prediction error below 3% in all cases.

The discovery that α_P is the master variable of ICL parallels Carnot's identification of T_c/T_h as the efficiency bound: a single measurable quantity determines the ceiling. Combined with Paper B's quality factor Q, the complete picture is the **Adaptation Cone Law**: on constant-ECC leaves, ICL occupies the compressive cone (requiring Q over-compensation) while fine-tuning occupies the expansive cone.

What determines α_P—the "Boltzmann question" of this framework—is the subject of Paper E.

References

[1] T. Brown et al., "Language models are few-shot learners," *NeurIPS*, 2020.

[2] H. R. Lee, "The d^2 pullback theorem for attention score manifolds," 2026.

[3] H. R. Lee, "The information theory of AI architectures: Effective computational capacity and intelligence decomposition," 2026.

[4] E. Akyürek, D. Schuurmans, J. Andreas, T. Ma, and D. Zhou, "What learning algorithm is in-context learning? Investigations with linear models," *ICLR*, 2023.

[5] S. M. Xie, A. Raghunathan, P. Liang, and T. Ma, "An explanation of in-context learning as implicit Bayesian inference," *ICLR*, 2022.

[6] J. von Oswald, E. Niklasson, E. Randazzo, et al., "Transformers learn in-context by gradient descent," *ICML*, 2023.

[7] E. Todd, M. L. Li, A. S. Sharma, and A. Mueller, "Function vectors in large language models," *ICLR*, 2024.

[8] R. Hendel, M. Geva, and A. Globerson, "In-context learning creates task vectors," *EMNLP Findings*, 2023.

[9] O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," *J. Multivariate Analysis*, 88(2):365–411, 2004.